

Reminders

Talk 7

• $X = \bigcup_{\lambda \in \Lambda} X_\lambda$ s.t.

• $D_c^b(X) =$ full subcat. of $D^b(\text{Sh } X)$ of constructible complexes

$\uparrow$
 $\mathcal{F} \in D^b(\text{Sh } X)$ is Λ -const. if all $H^i(\mathcal{F})$ are

• $D_\Lambda^b(X) =$ — " — Λ -const.

• $P_{D_\Lambda}^{\leq 0} := \{ \mathcal{F} \in D_{\text{constant}}^b(X_\lambda), H^i(\mathcal{F}) = 0, i > -d_\lambda \}$ $\mathcal{F} \mapsto H^i(\mathcal{F})[d_\lambda]$

• $P_{D_\Lambda}^{\geq 0} := \{ \text{— " — } i < -d_\lambda \}$ $P_{D_\Lambda}^{\leq 0} \cap P_{D_\Lambda}^{\geq 0} = \text{Loc}(X_\lambda)$

where $d_\lambda = \dim X_\lambda$

• $\mathcal{M}_\Lambda =$ heart of glued t-structures on all X_λ

• $\mathcal{M}_X =$ limit over all stratification $\Leftrightarrow \mathcal{F} \in \mathcal{M}_\Lambda$

($\mathcal{F}$ is perverse if it is Λ -perverse for some Λ)

Observation: "If X is smooth, then $\Lambda = \{*\}$, $X_* = X$, $\leadsto \mathcal{M}_X = \text{Loc}(X)$ "

"If Λ' refines Λ , then $\mathcal{M}_\Lambda \hookrightarrow \mathcal{M}_{\Lambda'}$ "

Claim: "Perverse sheaves behave like sheaves"

1) $\mathcal{M}_X$ is abelian;

2) If a morphism of perverse sheaves is 0 locally, then it is 0;

3) We can glue perverse sheaves over covers. $D = \text{cl. disk}$

Example $\mathcal{U} = \text{cl. unit disk}$ $x \in \partial D$ Goal: describe $\mathcal{F} \in \mathcal{M}_0$



1) $\mathcal{F}|_{\mathcal{U} \setminus \{0\}}$ is a local system, given by $(V, \mu) \xleftarrow{\text{monodromy}} \text{GL}(V)$
 $\uparrow$
 f.d. v.s.

Thm: $\exists$ equiv. of cats $\mathcal{M}_0 \Leftrightarrow \{ (V, V_0, \mu, a, b) : \begin{array}{l} V_0, V_0 \text{ v.s. } \mu \in \text{GL}(V) \\ a = V \rightarrow V_0, b = V_0 \rightarrow V \\ \text{s.t. } V \xrightarrow{\mu \cdot \text{id}} V \\ \begin{array}{ccc} a & \searrow & \nearrow b \\ & V_0 & \end{array} \end{array} \}$

Idea: $\mathcal{M}_0$ is k -linear + ab. $\Rightarrow$ study exact functors to nicer cat.

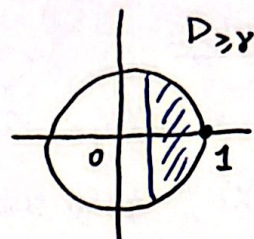
⚠ The stalk of $\mathcal{F}$ at D is a complex of v.s. concentrated in deg -1 & 0

• $H^0(\mathbb{F}_0)$ is only right exact

Write $D \setminus \{x\} \xrightarrow{\sim} D \xleftarrow{\sim} \{x\}$

Prop: $\mu_0 \rightarrow \text{Vect. } \mathbb{F} \mapsto H^0(D, \nu_! \nu^! \mathbb{F})$ is exact

concentrated in deg 0



for $|\gamma| \leq 1$

Study $H^*(D_{\geq \gamma}, \mathbb{F})$. If $0 < \gamma < 1$, $H^*(D_{\geq \gamma}, \mathbb{F}) = H^*(\mathbb{F}_1)$

$-1 < \gamma < 0$ $H^*(D_{\geq \gamma}, \mathbb{F}) = H^*(D, \mathbb{F})$

Note $D_{< \gamma} \xrightarrow{\hat{\nu}} D \xleftarrow{\hat{z}} D_{\geq \gamma} \Rightarrow$ LES

$\neq$ in coh. as we cross 0

$$\dots \rightarrow H^{-1}(D, \mathbb{F}_{D_{\geq \gamma}}) \rightarrow H^0(D, \hat{\nu}_! \hat{\nu}^! \mathbb{F}) \rightarrow H^0(D, \mathbb{F}) \rightarrow H^1(D, \mathbb{F}_{D_{\geq \gamma}}) \rightarrow \dots$$

$$H^0(D, \hat{\nu}_! \hat{\nu}^! \mathbb{F}) = H^0(D, \nu_! \nu^! \mathbb{F})$$

$$m \quad H^m(D, \nu_! \nu^! \mathbb{F}) \rightarrow H^m(D, \mathbb{F}) \rightarrow H^m(D, \mathbb{F}_x)$$

1	0	0	0
0	?	☆	0
-1	?	☆	☆
0	0	0	0

WTS: $H^{-1}(D, \mathbb{F}) \xrightarrow{\text{inj.}} H^1(D, \mathbb{F}_x)$. Apply the "Grand Octahedron"

$$\begin{array}{ccccc} \mathbb{Z} & \subset & \mathbb{Y} & \subset & D \\ \parallel & & \parallel & & \\ \{x\} & & S^1 & & \end{array}$$

$$\mathbb{F}_x[-1] \rightarrow (\mathbb{F}_{S^1})_{\nu_!}$$

$$\searrow \quad \uparrow \\ \mathbb{F}_{(D \setminus \{x\})!}$$

$$\sigma = \mu - \text{Id}$$

$$H^0 \rightarrow H^{-1}(D, \mathbb{F}_x) \xrightarrow{m} H^0_!(D, \mathbb{F}_{S^1 \setminus \{x\}})$$

$$\searrow \quad \uparrow \\ H^0(D, \nu_! \nu^! \mathbb{F})$$